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# SDC Hackathon

## Phase 1 : Preliminary Project Proposal

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Submission Deadline: 25 October 2026**

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

***Instagram:*** @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

**SDC HACKATHON 2026**

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## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | BIJAK MINDS |
| **Selected Domain** | Domain 2: SME Digital Operations, Workflow Automation & Compliance |
| **Selected SDG Track (1–17)** | **SDG 8: Decent Work and Economic Growth** *(Secondary: SDG 9 - Industry, Innovation & Infrastructure)* |
| **Project / Solution Name** | Kak Nadiah – Orbyte Smart SME Copilot |
| **Name (Leader)** | Ammar Hakim Bin Dzulhaidi |
| **Email & Phone (Leader)** | bijakminds@gmail.com 016-404 6294 |
| **University Affiliation(s)** | OPEN UNIVERSITY MALAYSIA |
| **Number of Team Members** | 1 |

## 2. Executive Summary & Problem Statement

### 2.1 The Problem Context

*Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.*

***[Maximum 250 words]***

In Southeast Asia, particularly Malaysia, micro and small-and-medium enterprises (MSMEs) form over 97% of business establishments. Despite their economic importance, grassroots business owners struggle with repetitive administrative bottlenecks, unauthorized supplier price markups, and impending mandatory e-invoicing compliance mandates (e.g., LHDN MyInvois).

Many small business operators (such as local grocers, food vendors, and retail merchants) receive unstructured paper receipts, WhatsApp invoices, and delivery orders. Cross-referencing billed line-items against negotiated contract rates is manual, slow, and prone to accounting leakages, draining up to 10–15 hours weekly per business owner. Furthermore, with mandatory digital taxation, conventional enterprise ERP software is too costly, rigid, and complex for micro-enterprises.

This creates operational fragility, reduced profit margins, and non-compliance risk. Without an accessible, localized, and intelligent interface, MSMEs risk falling behind the national digital economy. Solving this matters because protecting SME cash flow and streamlining legal digital workflows directly protects local livelihoods, promotes inclusive economic growth, and accelerates sustainable digital transformation.

### 2.2 The Proposed AI Solution

*Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.*

***[Maximum 300 words]***

**Kak Nadiah** is a localized, multimodal Smart SME AI Copilot engineered specifically for grassroots enterprise workflows. Designed with a clean Neo-Brutalism interface, it bridges the gap between chaotic physical receipts and structured enterprise compliance through three core autonomous modules:

1. **AutoProcure & Discrepancy Detection:** Utilizing Google Gemini 2.5 Flash Vision, Kak Nadiah extracts line-items from invoice photos or PDF files, matching claimed unit prices against stored contract benchmarks. Any unauthorized price discrepancy (e.g., unexpected raw material price hikes) is automatically flagged with actionable summaries.
2. **Compliant e-Invoice & Formal Letter Engine:** Automates standard LHDN-compliant tax invoice generation featuring validation QR codes, as well as customizable formal administrative

letters integrated with locked master company assets (logo and signature stamps) and exportable directly to A4 PDF, email (Resend API), or WhatsApp.

3. **Voice Executive Assistant & Multi-Language Support:** Powered by speech recognition, Web Speech synthesis / ElevenLabs, and on-demand UI translation supporting over 249+ languages, allowing operators of diverse backgrounds to converse naturally via voice or text.
4. **Human-in-the-Loop & Security Architecture:** No financial disbursement or ledger synchronization occurs autonomously without employer authorization enforced via Google Authenticator (TOTP 2FA).

Leveraging skills from SDC Phase 1 workshops, the application combines Gemini multimodal APIs, Express.js microservices, persistent local/server JSON storage, and seamless simulated synchronization into the Orbyte Cloud ecosystem.

## 3. Technical Architecture & Implementation Plan

### 3.1 Conceptual Tech Stack

*List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.*

*Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.*

**Frontend Architecture:**

- **UI & Styling:** HTML5, Tailwind CSS (Neo-Brutalism Design System), Vanilla JavaScript, FontAwesome 6, Plus Jakarta Sans typography.
- **Data Visualization:** Chart.js (Real-time operational expense analytics & trend tracking).
- **Internationalization (i18n):** Hybrid dual-layer translation engine (Client-side instant cache for 11 core languages + Google Gemini on-demand translation supporting 249+ languages & browser-native translation compatibility).
- **Audio & Document Utilities:** Web Speech API, client-side base64 canvas/file processors, and browser-native A4 print-to-PDF formatting engine.

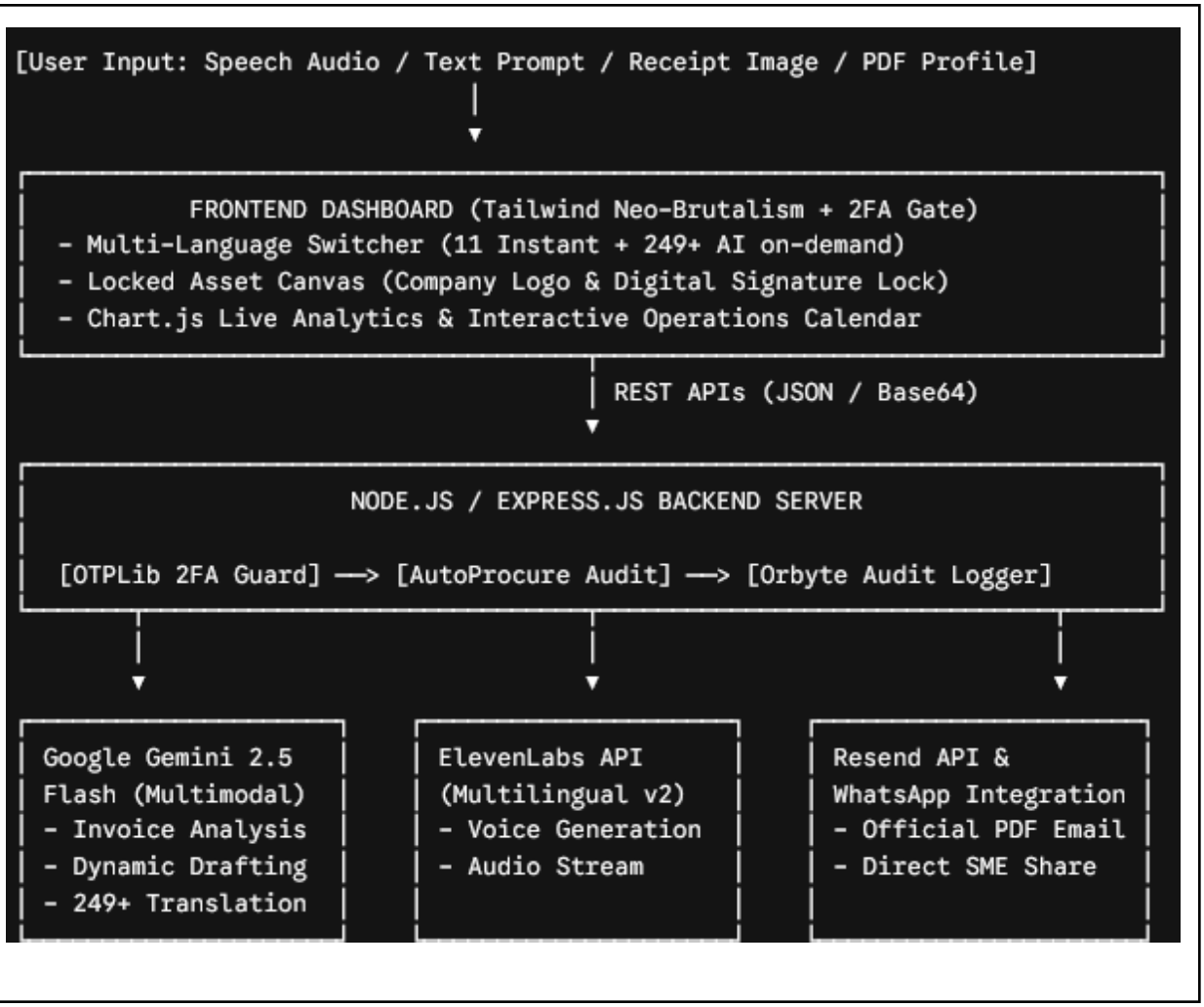
**Backend & Cloud Microservices:**

- **Runtime & Framework:** Node.js & Express.js REST API server hosted on containerized cloud infrastructure (Replit).
- **Security & 2FA Governance:** OTPLib (RFC 6238 TOTP Google Authenticator 2FA) and QRCode rendering library for strict Human-in-the-Loop authorization.
- **Multichannel Communication & Delivery: Resend API** for automated transactional official letter dispatch to Gmail/Outlook, plus direct WhatsApp Web routing.
- **Data & Audit Persistence Layer:** Persistent JSON document store (`chat_history.json`, master company profile registry, and Orbyte workspace audit logs).

**AI Models & Third-Party APIs Utilized:**

- **Google Gemini 2.5 Flash API (Google Cloud / AI Studio - SDC Sponsor Platform):** Multimodal document & receipt parsing (Vision), discrepancy detection, automated LHDN e-invoicing classification, dynamic official letter drafting, and multilingual translation.
- **ElevenLabs Multilingual v2 API:** Neural Voice AI synthesis delivering natural, expressive Bahasa Melayu audio playback for Kak Nadiah Voice Copilot.

**System Architecture & Data Flow Diagram:**




### 3.2 Design-Thinking & Deployment Strategy

*Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.*

***[Maximum 200 words]***

We applied design thinking by empathizing directly with non-tech-savvy SME traders who find standard accounting software intimidating.

- **Empathize & Define:** Micro-merchants operate on the go and lack bookkeeping time. They require frictionless inputs (camera snapshots and voice commands) rather than dense multi-column forms.
- **Ideate & Prototype:** We created "Kak Nadiah"—a friendly, authoritative AI persona. The Neo-Brutalism UI employs bold contrast, tactile borders, clear color coding (green for matched, red for price hikes), and multi-language inclusivity (supporting Malay, English, Mandarin, Tamil, and 249+ global dialects) to eliminate language barriers.

- **Test & Deploy:** By keeping the human in the loop via mandatory 2FA Google Authenticator, business owners retain total authority over payments. For Phase 2, the containerized Node.js stack is optimized for low-latency edge deployment, lightweight resource consumption, and seamless webhook integration with live accounting ERPs and LHDN MyInvois API sandboxes.

## 4. Preliminary Environmental & SDG Impact Framework

*To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.*

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess* ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

| **Metric Category** | **Your Response** |
|---|---|
| **Target SDG Sub-Target(s)**<br>*Identify the exact UN SDG sub-target your project addresses*<br>*e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."* | **Target 8.3:** *"Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services."* |
| **Current Problem Baseline**<br>*What is the starting condition of the problem, expressed as a number?*<br>*e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."* | MSMEs currently spend an estimated **12 to 15 administrative hours weekly** on manual receipt sorting and supplier reconciliation, while suffering an average **3% to 5% invoice leakage** from undetected vendor price discrepancies and manual clerical errors. |
| **Core Measurable KPI**<br>*The specific unit you will track*<br>*e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.* | **1. Administrative Processing Time Reduction (%):** Time taken to process, audit, and log an invoice or formal letter (benchmark: minutes vs. seconds).<br>**2. Invoice Discrepancy Detection Accuracy Rate (%):** Precision in flagging unauthorized price markups. |

| Metric Category | Your Response |
|---|---|
| | **3. Paper Waste Reduction (kg / sheets avoided):** Complete digitization of thermal receipts and paper forms. |
| **Phase 2 Impact Target**<br>*Your realistic, percentage-based target by the final prototype presentation*<br>*e.g., "20% improvement in computational resource use."* | **80% reduction in invoice audit cycle time** (from 15 minutes manual cross-check down to under 5 seconds per invoice), **zero undetected billing overcharges** within standard inventory items, and **100% digital compliance readiness** for mandated e-invoicing. |

## 5. Eligibility & Readiness Checklist

*Tick all boxes that apply before submitting.*

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

## 6. IP & Originality Declaration

### 6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

### 6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

| Name | Type (code/dataset/model) | License |
|---|---|---|

### 6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

## 7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[24 October 2026]